

Kavya Byju

SOC Analyst | Threat Intelligence | Vulnerability Management | Penetration Testing

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PROFESSIONAL SUMMARY

Cybersecurity graduate with hands-on experience in threat intelligence, vulnerability assessment, penetration testing, and security monitoring. Built an AI-powered threat investigation platform integrating multiple threat intelligence APIs and MITRE ATT&CK mapping. Proficient in Python, FastAPI, Splunk, ELK Stack, and Active Directory security. Seeking an entry-level SOC Analyst, Security Analyst, or Junior Penetration Tester role.

PROJECTS

PHANTOM.TRACE – AI-Powered Threat Investigation Platform | Personal, 04/2026 – Present
Python • FastAPI • VirusTotal API • Gemini AI • MITRE ATT&CK • GitHub

- Built a multi-source threat intelligence ingestion platform integrating VirusTotal, Shodan, and AbuseIPDB APIs with automated threat scoring (Low/Medium/High/Critical).
- Developed a FastAPI backend to aggregate, parse, and normalize security event data for structured analysis.
- Integrated MITRE ATT&CK mapping and AI-generated analyst explanations for each investigation result.
- Deployed a live web dashboard UI enabling one-click PDF incident report generation.

LLM-Assisted Threat Intelligence & Incident Analysis | Capstone | Seneca Polytechnic, 01/2026 – 04/2026
Python • RAG • Qwen 3.5 • ChromaDB • OpenAI Embeddings • ELK Stack • MITRE ATT&CK

- Engineered a centralized CTI knowledge base unifying 9,000+ intelligence records from MITRE ATT&CK, MISP Galaxy, IoC repositories, and APT reports.
- Built Python ETL pipelines to collect, clean, normalize, and export structured threat data for semantic retrieval and cross-source correlation.
- Implemented a RAG pipeline (OpenAI embeddings + ChromaDB + Qwen 3.5) enabling natural-language analyst queries across threat actors and ATT&CK techniques.
- Deployed ELK Stack on AWS with Filebeat log shipping and Kibana dashboards for SOC-style ATT&CK coverage analysis.

Active Directory & Infrastructure Hardening | Seneca Polytechnic, 12/2025 – 12/2025

- *Windows Server 2016 • Kali Linux • Nessus • Impacket • Nmap • Enum4linux*
- Configured Windows Server 2016 as Domain Controller, performed network reconnaissance, lateral movement simulation, and privilege escalation from Kali Linux.
- Applied least-privilege access controls and Group Policy hardening, then validated mitigations by retesting all attack paths.
- Documented attack vectors, privilege escalation paths, and remediation steps in an enterprise-style security assessment report.

Vulnerability Assessment & Web Exploitation Lab, 11/2025
Seneca Polytechnic | Nikto, Metasploit, Drupalgeddon2 (CVE-2018-7600)

- Conducted web vulnerability scanning with Nikto, then exploited a vulnerable Drupal target using Metasploit's Drupalgeddon2 module.
- Established a remote Meterpreter shell and performed post-exploitation analysis to assess attacker access scope.
- Authored a formal vulnerability report with CVSS-informed risk ratings and step-by-step remediation guidance.

WORK EXPERIENCE

Bakery Associate and Cashier, T&T Bakery and Cafe

09/2023 – Present
Markham

- Managed high-volume, time-sensitive customer service operations, building composure and rapid decision-making skills under pressure.
- Maintained near-zero transaction error rates across thousands of orders, demonstrating precision and attention to detail.

Computer Lab / IT Support Assistant, Micron Computers

05/2023 – 08/2023
India

- Assisted students and staff with hardware, software, and network connectivity troubleshooting.
- Installed, configured, and updated software applications on desktop and laptop computers.
- Performed password resets and resolved account access issues for employees and customers.
- Maintained printers and peripherals; guided users on effective use of computer systems and applications.

EDUCATION

Cybersecurity & Threat Management, (Ontario College Graduate Certificate), Seneca Polytechnic

09/2025 – 04/2026
Toronto, ON

Computer Programming (Ontario College Diploma), Seneca Polytechnic

09/2023 – 08/2025
Toronto, ON

SKILLS

Offensive Security: Metasploit, Nmap, Burp Suite, Nikto, Privilege Escalation (SUID/SGID), Drupalgeddon2, Meterpreter

Defensive & Monitoring: Splunk, ELK Stack, Wazuh, Suricata IDS, Wireshark, Nessus, Log Analysis, Alert Triage

Threat Intelligence: MITRE ATT&CK, MISP Galaxy, VirusTotal, Shodan, AbuseIPDB, IOC Analysis, OSINT, RAG, ChromaDB

Cloud & IAM: AWS (EC2, VPC, IAM), GCP IAM, OAuth 2.0, Active Directory, RBAC, Least Privilege, Group Policy

Web & App Security: OWASP Top 10, SQLi, XSS, Web Vulnerability Analysis

Programming: Python, FastAPI, Bash, PowerShell, JavaScript, SQL, REST APIs, Git/GitHub

Frameworks: MITRE ATT&CK, NIST CSF, CIS Controls, CVSS v3.1, Incident Response Lifecycle

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

OPSWAT – Introduction to Critical Infrastructure Protection (ICIP)

AWS Academy Graduate – Cloud Security Foundations

Introduction to Cybersecurity

C5W-100 – Introduction to Digital Forensics

Introduction to the EASY Framework for Threat Intelligence

LFS101: Introduction to Linux

Self-Directed Cybersecurity Training(TryHackMe, Hack The Box)

Microsoft Certified: Security, Compliance, and Identity Fundamentals

Volunteer – ION Iconic 2026

Supported event operations and assisted with attendee coordination at ION's annual flagship conference, gaining firsthand exposure to ION's product ecosystem, client network, and company culture

Certification Preparation – CompTIA Security+ & Microsoft SC-900

Actively studying Security+ domains (threats, cryptography, network security, IAM, risk management).